

Data Engineering Mid-Semester Project

Project Overview

Design and present a data warehouse solution for a business of your choice.

You can choose a topic related to one of your previous / current projects or choose a completely new topic. No duplicated/similar topics allowed. This project will test your understanding of data modeling, pipeline design, and modern data architecture patterns.

Choose when to present your work:  When do you present?

Please add the link to your slides week prior to the presentation date so I will have enough time to generate questions. If you won't add the presentation week prior, your slot will be moved to the next available one.

Technical Requirements

Data Sources (Minimum 3)

1. At least one streaming data source
 - Real-time data processing
 - Event-driven architecture
 - Examples: IoT sensors, user activity, real-time transactions
2. At least one source with late arrivals
 - Data arriving up to 48 hours after event time
 - Must handle out-of-order data
 - Example: Third-party analytics, partner data feeds
3. Additional data source(s)
 - Batch processing from another data mart
 - Third party API integration
 - File-based imports

Data Modeling Requirements

1. At least one static dimension table
 - Example: Product catalog, location master, category reference
2. Slowly Changing Dimension (SCD)
 - At least one Type 2 SCD
 - Must track historical changes
 - Example: Customer preferences, pricing tiers

Architecture Requirements

1. Bronze Layer
 - Raw data ingestion
 - Schema documentation
2. Silver Layer
 - Standardization processes
 - Join and enrichment logic
3. Gold Layer
 - Business metrics
 - Aggregation strategies
 - End-user views

Business Requirements

1. Business Analytics Requirements
 - Define a key business question
 - Design a mock dashboard to answer this question
 - Backtrack to design fact/dimension table(s) that will feed this dashboard
2. Machine Learning Requirements
 - Define a business problem suitable for ML
 - Design a dataset/feature table for this ML model
 - Backtrack to design fact/dimension table(s) needed for this dataset

Presentation Requirements

Business Context

- Industry overview & Business problem
- Key stakeholders who will use the data
- Data sources explanation
- Dashboard to support
- Dataset creation for ML model

Technical Design

- Architecture diagram
- Table schemas (Bronze, Silver, Gold, Facts & Dimensions)
- Data flow diagrams
- How the gold layer will produce the dashboard and ML model

If you need to show off additional tables or dashboards, add screenshots of them to your slides with links to full view.

Evaluation Criteria

Data Design (30%)

Individual Presentation Abilities (20%)

You need to sound like you truly contributed to the project.

Question Response (30%)

I will ask 2-3 questions, depending on the team's size. Each student needs to answer a single question, the response will be graded individually.

Peer Feedback (20%)

- Constructive criticism about data modeling of another team (10%)
- Improvement suggestions regarding data modeling of another team (10%)

Submission Guidelines

Presentation Format 📅 When do you present?

- 10-12 minute presentation
- 5-10 minute Q&A
- Link to slides

Team Requirements

- 2-3 members per team
- **Individual** Q&A responses

Additional Notes

- Focus on real-world applicability
- Consider scalability
- Document assumptions
- Prepare for edge cases